

WASP-43b

R-2731

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_120314 · created 2026-09-02T00:23:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456000.68415 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.021
Transit depth (Rp/Rs)^2	3.92 +/- 0.83 [%]
Orbital Inclination (inc)	83.86 +/- 1.81
Airmass coefficient 1 (a1)	0.935 +/- 0.034
Airmass coefficient 2 (a2)	0.05 +/- 0.02
Scatter in the residuals of the lightcurve fit is	3.68 %
Best Comparison Star	#3 - [99, 244]
Optimal Aperture	3.42
Optimal Annulus	4.61
Transit Duration (day)	0.0572 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.021 vs 0.1594 (deviation 1.62 σ)
Duration fit vs published	0.0572 d vs 0.0512 d (deviation 1.17 σ)
Mid-time O–C	0.6 min
Depth SNR	8.3
Residual scatter	3.68 %

Light curve

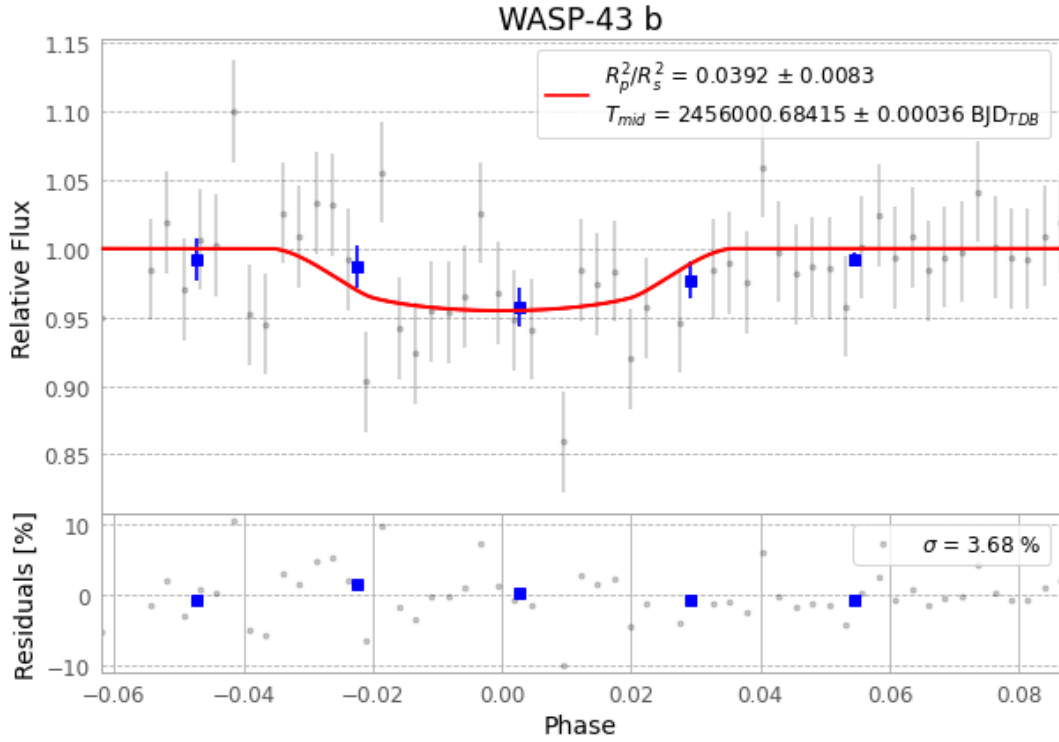


Figure 1 — FinalLightCurve_WASP-43 b_2012-03-13.png (SHA-256 a5c11248acc67add...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [165, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6df9640714c34ef6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), WASP-43120314030314.FITS → WASP-43120314055712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-120314.log (8dde8babeeab...), FinalLightCurve_WASP-43 b_2012-03-13.png (a5c11248acc6...), FinalLightCurve_WASP-43 b_2012-03-13.csv (8c7692d28c00...)

Compute resources

Wall time 121.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:23Z.